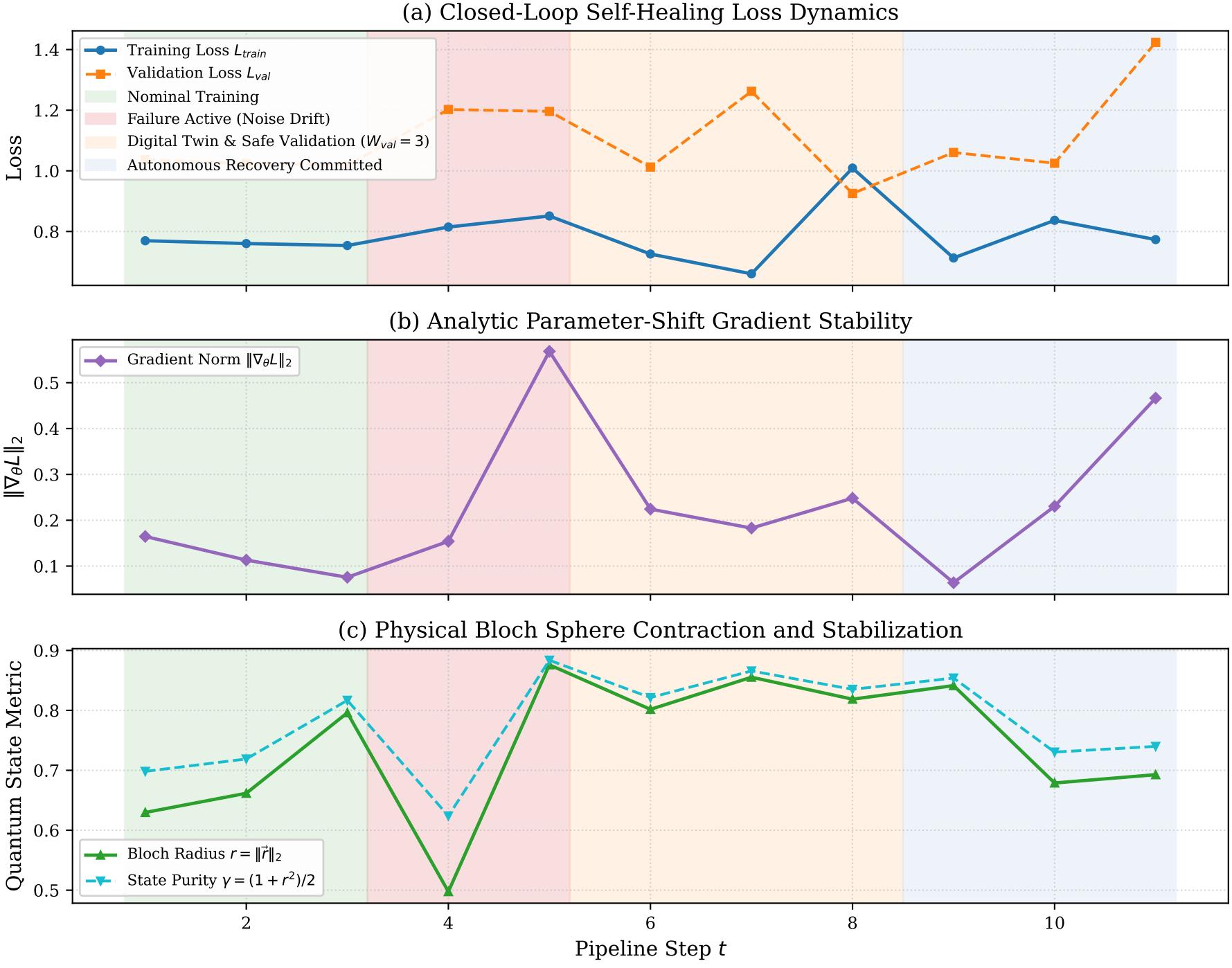
(a) Closed-Loop Self-Healing Loss Dynamics
Loss
Training Loss L_{train}
Validation Loss L_{val}
Nominal Training
Failure Active (Noise Drift)
Digital Twin & Safe Validation ($W_{val} = 3$)
Autonomous Recovery Committed
1.4
1.2
1.0
0.8
(b) Analytic Parameter-Shift Gradient Stability
$\|\nabla_\theta L\|_2$
Gradient Norm $\|\nabla_\theta L\|_2$
0.5
0.4
0.3
0.2
0.1
(c) Physical Bloch Sphere Contraction and Stabilization
Quantum State Metric
Bloch Radius $r = \|\vec{r}\|_2$
State Purity $\gamma = (1 + r^2)/2$
0.9
0.8
0.7
0.6
0.5
2
4
6
8
10
Pipeline Step t